

Basic Concepts of Probability

- Sample space = **sure event**, subset of $S = \varnothing$ = **null event**
- Mutually exclusive/disjoint** if $A \cap B = \varnothing$
- Contained:** $A \subset B \equiv B \supset A$.
- If $A \subset B$ and $B \supset A$, then $A = B$

Basic Properties

- $(A \cap B)' = A' \cup B'$
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $(A \cup B)' = A' \cap B'$
- $A \cup B = A \cup (B \cap A')$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A = (A \cap B) \cup (A \cap B')$

De Morgan’s Law

- $(\bigcup_{r=1}^n A_r)' = \bigcap_{r=1}^n (A_r)'$
- $(\bigcap_{r=1}^n A_r)' = \bigcup_{r=1}^n (A_r)'$

Counting Methods

Multiplication & Addition Principle

Permutation

- An arrangement of r objects from a set of n objects, $r \leq n$, order taken into consideration.
- n distinct objects taken r at a time = $nPr = \frac{n!}{(n-r)!}$
- In a circle: $(n-1)!$
- Not all are distinct: $\sum_{r=1}^k n_k = n$, $nPr_1, n_2, ..., n_k = \frac{n!}{n_1!n_2!...n_k!}$

Combination

- No of ways selecting r from n objects w/o regarding order
- $\binom{n}{r} = {}_n C_r = \frac{n!}{r!(n-r)!}$, ${}_n C_r \times r! = {}_n P_r$
- $\binom{n}{r}$ = binom coeff of the term $a^r b^{n-r}$ in binom expansion of $(a+b)^n$:
 - $\binom{n}{r} = \binom{n}{n-r}$ for $r = 0, 1, ..., n$
 - $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ for $1 \leq r \leq n$
 - $\binom{n}{r} = 0$ for $r < 0$ pr $r > n$

Relative frequency (f_A)

$f_A = \frac{n_A}{n}$, event A in n repetitions of experiment E , n_A = no of times that event A occurred among the n repetitions.

Axioms of Probability

- If $A_1, A_2, ...$ are mutually exclusive (disjoint), i.e. $A_i \cap A_j = \varnothing$ when $i \neq j$, then $\Pr(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \Pr(A_i)$
If events A and B are mutually exclusive, then $\Pr(A \cup B) = \Pr(A) + \Pr(B)$

Properties of Probability

- If $A_1, A_2, ..., A_n$ are mutually exclusive, then $\Pr(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \Pr(A_i)$
- $\Pr(A) = \Pr(A \cap B) + \Pr(A \cap B')$
- $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$
- $\Pr(A \cup B \cup C) = \Pr(A) + \Pr(B) + \Pr(C) - \Pr(A \cap B) - \Pr(B \cap C) - \Pr(A \cap C) + \Pr(A \cap B \cap C)$
- The Inclusion-Exclusion Principle**

$$\Pr(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \Pr(A_i) - \sum_{i=1}^{n-1} \sum_{j=i+1}^n \Pr(A_i \cap A_j) + \sum_{i=1}^{n-2} \sum_{j=i+1}^{n-1} \sum_{k=j+1}^n \Pr(A_i \cap A_j \cap A_k) - ...$$

Conditional Probability, $P(A \mid B)$

- $\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$, if $\Pr(A) \neq 0$
- For fixed A , $\Pr(B \mid A)$ satisfies the postulates of probability.
- False positive: $\Pr(+ \mid \text{condition})$

Multiplication rule

- $\Pr(A \cap B) = \Pr(A) \Pr(B \mid A) = \Pr(B) \Pr(A \mid B)$, providing $\Pr(A) > 0, \Pr(B) > 0$
- $\Pr(A \cap B \cap C) = \Pr(A) \Pr(B \mid A) \Pr(C \mid A \cap B)$
- $\Pr(A_1 \cap ... \cap A_n) = \Pr(A_1) \Pr(A_2 \mid A_1) \Pr(A_3 \mid A_1 \cap A_2) ... \Pr(A_n \mid A_1 \cap ... \cap A_{n-1})$

The Law of Total Probability

- Let $A_1, A_2, ..., A_n$ be a partition of sample space S (mutually exclusive & exhaustive events s.t. $A_i \cap A_j = \varnothing$ for $i \neq j$ and $\bigcup_{i=1}^n A_i = S$).
- Then $\Pr(B) = \sum_{i=1}^n \Pr(B \cap A_i) = \sum_{i=1}^n \Pr(A_i) \Pr(B \mid A_i)$

Bayes’ Theorem

- Let $A_1, A_2, ..., A_n$ be a partition of S
- $\Pr(A_k \mid B) = \frac{\Pr(A_k) \Pr(B \mid A_k)}{\sum_{i=1}^n \Pr(A_i) \Pr(B \mid A_i)} = \frac{\Pr(A_k) \Pr(B \mid A_k)}{\Pr(B)}$, $k \in [1, n]$

Independent Events

- Definition: iff $\Pr(A \cap B) = \Pr(A) \Pr(B)$

Properties

- Suppose $\Pr(A) > 0, \Pr(B) > 0$, A and B are independent:
 - $\Pr(B \mid A) = \Pr(B)$ and $\Pr(A \mid B) = \Pr(A)$
 - A and B cannot be mutually exclusive (and vice versa)
- The sample space S and $\varnothing$ are independent of any event
- If $A \subset B$, then A and B are dependent unless $B = S$
Warning: Indep events can’t be shown using Venn Diagram, so calc!!!

Theorem

If A, B are indep, then so are A and B' , A' and B , A' and B' .

n Independent Events

- Pairwise Independent Events:**
Events $A_1, A_2, ..., A_n$ are pairwise indep iff $\Pr(A_i \cap A_j) = \Pr(A_i) \Pr(A_j)$
- Mutually Independent:**
Events $A_1, A_2, ..., A_n$ are (mutually) independent iff for any subset $\{A_{i_1}, A_{i_2}, ..., A_{i_k}\}$ of $A_1, A_2, ..., A_n$, $\Pr(A_{i_1} \cap A_{i_2} \cap ... \cap A_{i_k}) = \Pr(A_{i_1}) \Pr(A_{i_2}) ... \Pr(A_{i_k})$

Remarks

- $A_1, A_2, ..., A_n$ are mutually independent $\Leftrightarrow$ for any pair of events A_j, A_k where $j \neq k$, the multiplication rule holds, for any 3 distinct events, the multiplication rule holds, and so on $\Pr(A_1 \cap A_2 \cap ... \cap A_n) = \Pr(A_1) \Pr(A_2) ... \Pr(A_n)$. In total there are $2^n - n - 1$ diff cases.
- Mutually indep $\Rightarrow$ pairwise indep (not the converse)
- Suppose $A_1, A_2, ..., A_n$ are mutually indep events, let $B_i = A_i$ or A'_i , $i \in [1, n]$. Then $B_1, B_2, ..., B_n$ are also mutually indep events.

Concepts of Random Variables

Equivalent Events

Definition

- Let E be an experiment in sample space S . Let X be an R.V. defined on S , and R_X its range space, i.e. $X : S \rightarrow \mathbb{R}$
- Let B be an event w.r.t. R_X , i.e. $B \subset R_X$
- Suppose $A = \{s \in S \mid X(s) \in B\}$ (A consists of all sample points s in S for which $X(s) \in B$)
- A and B are **equivalent events**, and $\Pr(B) = \Pr(A)$

Example

- Consider tossing a coin twice, $S = \{HH, HT, TH, TT\}$
- Let X be no of heads, then $R_X = \{0, 1, 2\}$
- $A_1 = \{HH\}$ equiv $B_1 = \{2\}$, $A_2 = \{HT, TH\}$ equiv $B_2 = \{1\}$, $A_3 = \{TT\}$ equiv $B_3 = \{0\}$, $A_4 = \{HH, HT, TH\}$ equiv $B_4 = \{2, 1\}$

Discrete Probability Distributions

Discrete R.V.

Let X be an RV. If R_X is **finite or countable infinite**, X is discrete RV

Probability Fn (p.f.) or Probability Mass Function (p.m.f.)

- For a discrete R.V., each value X has a certain probability $f(x)$. Such a function $f(x)$ is called the p.f.
- The collection of pairs $(x_i, f(x_i))$ is prob distribution of X
- The probability of $X = x_i$ denoted by $f(x_i)$ must satisfy: $f(x_i) \geq 0 \forall x_i$ and $\sum_{i=1}^{\infty} f(x_i) = 1$

Continuous Probability Distributions

Continuous R.V.

Suppose that R_X is an **interval or a collection of intervals**, then X is a continuous R.V.

Probability Density Function (p.d.f.)

- Let X be a continuous R.V.
- p.d.f. $f(x)$ is a function satisfying:
 - $f(x) \geq 0 \forall x \in R_X$
 - $\int_{R_X} f(x) dx = 1$ or $\int_{-\infty}^{\infty} f(x) dx = 1$ as $f(x) = 0 \forall x \notin R_X$
 - $\forall c, d : c < d$ (i.e. $(c, d) \subset R_X$), $\Pr(c \leq X \leq d) = \int_c^d f(x) dx$

Remarks

- $\Pr(c \leq X \leq d) = \int_c^d f(x) dx$ represents area under the graph of the p.d.f. $f(x)$ between $x = c$ and $x = d$
- Let x_0 be a fixed value, $\Pr(X = x_0) = 0$
- $\leq$ and $<$ can be used interchangeably in a prob statement.
- $\Pr(A) = 0$ does not necessarily imply $A = \varnothing$
- $R_X \in [a, b] \Rightarrow f(x) = 0 \forall x \notin [a, b]$

Cumulative Distribution Function (c.d.f.)

Let X be an R.V., disc or cont. $F(x)$ is a cdf of X where $F(x) = \Pr(X \leq x)$

c.d.f. for Discrete R.V.

- $F(x) = \Pr(\sum_{t \leq x} f(t) = \sum_{t \leq x} \Pr(X = t))$
- c.d.f. of a discrete R.V. is a step function
- $\forall a, b$ s.t. $a \leq b$, $\Pr(a \leq X \leq b) = \Pr(X \leq b) - \Pr(X < a) = F(b) - F(a^-)$ where a^- is the largest possible value of X strictly less than a
- $R_X \subset \mathbb{Z}, a, b \in \mathbb{Z} \Rightarrow$
 - $\Pr(a \leq X \leq b) = \Pr(X = a \text{ or } a + 1 \text{ or } ... \text{ or } b) = F(b) - F(a - 1)$
 - Taking $a = b$, $\Pr(X = a) = F(a) - F(a - 1)$

c.d.f. for Continuous R.V.

- $F(x) = \int_{-\infty}^{\infty} f(t) dt$
- $f(x) = \frac{dF(x)}{dx}$ if the derivative exists
- $\Pr(a \leq X \leq b) = \Pr(a < X \leq b) = F(b) - F(a)$
- $F(x)$ is a non-decreasing function: $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$; and $0 \leq F(x) \leq 1$

Mean and Variance of an R.V.

Expected Value / Mean / Mathematical Expectation

- Discrete:** $E(X) = \mu_X = \sum_i x_i f(x_i) = \sum_x x f(x)$
- If $f(x) = \frac{1}{N}$ for each of the N values of x , $E(X) = \frac{1}{N} \sum_i x_i$
- Continuous:** $E(X) = \mu_X = \int_{-\infty}^{\infty} x f(x) dx$
- Remark:** The expected value exists if the sum/integral exists

Expectation of a function of an R.V.

$\forall g(X)$ with p.f. $f_X(x)$

- Discrete:** $E[g(X)] = \sum_x g(x) f_X(x)$
- Continuous:** $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$
- Provided the sum/integral exists.

2.5.3 Variance ($\sigma_X^2 = V(X)$)

- $g(x) = (x - \mu_X)^2$, Let X be an R.V. with p.f. $f(x)$
- $\sigma_X^2 = V(X) = E[(X - \mu_X)^2]$
- $E[(X - \mu_X)^2] = \begin{cases} \sum_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$
- $V(X) \geq 0$, $V(X) = E(X^2) - [E(X)]^2$
- Standard deviation** $= \sigma_X = \sqrt{V(X)}$

2.5.4 K-th moment of X

- Definition:** $E(X^k)$, use $g(x) = x^k$ in expectation of a fn

2.5.5 Properties of Expectation

- $E(aX + b) = aE(X) + b$
- $V(X) = E(X^2) - [E(X)]^2$
- $V(aX + b) = a^2 V(X)$

2.6 Chebyshev's Inequality

- Let X be an R.V. with $E(X) = \mu$, $V(X) = \sigma^2$
- $\forall k > 0$, $\Pr(|X - \mu| > k\sigma) \leq \frac{1}{k^2}$ OR $\Pr(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$
- Holds for **all** distributions with finite mean and variance
- Gives a **lower bound** but not exact probability.

3 2D RV & Conditional Probability Distributions

3.1 2D RV Definition (Random Vector)

- Let E be experiment and S sample space assoc with E . Let X and Y be 2 functions each assigning a real number to each $s \in S$. (X, Y) is a 2D RV
- Range Space:** $R_{X,Y} = \{(x,y) \mid x = X(s), y = Y(s), s \in S\}$
- The definition can be extended to n -dimensional RV (or n -dimensional random vector) for $X_1, X_2, \dots, X_n$.
- (X, Y) is a 2D discrete RV if the possible values of $(X(s), Y(s))$ are **finite** or **countable infinite**.
- (X, Y) is a 2D continuous RV if the possible values of $(X(s), Y(s))$ can **assume all values in some region** of the Euclidean plane $\mathbb{R}^2$

3.2 Joint Probability Density Function

3.2.1 For Discrete RV

Let (X, Y) be a 2D discrete RV. With each possible value (x_i, y_j) , we associate a number $f_{X,Y}(x_i, y_j)$ representing $\Pr(X = x_i, Y = y_j)$ and satisfying:

- $f_{X,Y}(x_i, y_j) \geq 0 \forall (x_i, y_j) \in R_{X,Y}$
- $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} f_{X,Y}(x_i, y_j) = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \Pr(X = x_i, Y = y_j) = 1$

The function $f_{X,Y}(x, y)$ defined $\forall (x_i, y_j) \in R_{X,Y}$ is called **joint probability function** of (X, Y) .

Let A be any set consisting of pairs of (x, y) values, then:

$$\Pr((X, Y) \in A) = \sum \sum_{(x,y) \in A} f_{X,Y}(x, y)$$

3.2.2 For Continuous RV

Let (X, Y) be a 2D continuous RV assuming all values in some region R of the Euclidean plane $\mathbb{R}^2$. $f_{X,Y}(x, y)$ is called joint pdf if it satisfies:

- $f_{X,Y}(x, y) \geq 0 \forall (x, y) \in R_{X,Y}$
- $\iint_{(x,y) \in R_{X,Y}} f_{X,Y} dy dx = 1$ or $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy dx = 1$

3.3 Marginal and Conditional Probability Distributions

3.3.1 Marginal Probability Distributions

Let (X, Y) be a 2D RV with joint pdf $f_{X,Y}(x, y)$. The **marginal probability distributions** of X and Y are:

- Discrete:** $f_X(x) = \sum_y f_{X,Y}(x, y)$ and $f_Y(y) = \sum_x f_{X,Y}(x, y)$
- Cont:** $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$

3.3.2 Conditional Distribution

Let (X, Y) be a 2D RV with joint pdf $f_{X,Y}(x, y)$, let $f_X(x)$ and $f_Y(y)$ be the marginal probability functions of X and Y respectively. Then the **conditional distribution of Y given that $X = x$:**

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}, \text{ if } f_X(x) > 0 \text{ for each } x \text{ in range of } X$$

Similarly, the **conditional distrubution of X given $Y = y$:**

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}, \text{ if } f_Y(y) > 0 \text{ for each } y \text{ in range of } Y$$

Remarks:

- The conditional pdf satisfy all the reqs for a 1D pdf:
 - For a fixed y , $f_{X|Y}(x \mid y) \geq 0$, for a fixed x , $f_{Y|X}(y \mid x) \geq 0$
 - For discrete RV: $\sum_x f_{X|Y}(x \mid y) = 1$ and $\sum_y f_{Y|X}(y \mid x) = 1$
 - For cont RV: $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y) dx = 1$ and $\int_{-\infty}^{\infty} f_{Y|X}(y \mid x) dy = 1$
- For $f_X(x) > 0$, $f_{X,Y}(x, y) = f_{Y|X}(y \mid x)f_X(x)$. For $f_Y(y) > 0$, $f_{X,Y}(x, y) = f_{X|Y}(x \mid y)f_Y(y)$

3.4 Independent RV

RV X and Y are independent iff $f_{X,Y}(x, y) = f_X(x)f_Y(y) \forall x, y$
This defn can be extended to RV $X_1, X_2, \dots, X_n$

- The product of 2 positive functions $f_X(x)$ and $f_Y(y)$ means a function which is positive on a **product space**.
- i.e. if $f_X(x) > 0$ for $x \in A_1$ and $f_Y(y) > 0$ for $x \in A_2$, then $f_X(x)f_Y(y) > 0$ for $(x, y) \in A_1 \times A_2$

3.5 Expectation

$$E[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y) f_{X,Y}(x, y) & \text{for Disc RV} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy & \text{for Cont RV} \end{cases}$$

3.5.1 Covariance

Let $g(X, Y) = (X - \mu_X)(Y - \mu_Y)$.
Let (X, Y) be a bivariate RV with joint pdf $f_{X,Y}(x, y)$, then the **covariance** of X, Y is $Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$

- Discrete:** $Cov(X, Y) = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y)$
- Cont:** $Cov(X, Y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{X,Y}(x, y) dx dy$
- $Cov(X, Y) = E(XY) - \mu_X \mu_Y$
- If X, Y are independent, then $Cov(X, Y) = 0$.
- $Cov(aX + b, cY + d) = acCov(X, Y)$
- $V(aX + bY) = a^2 V(X) + b^2 V(Y) + 2abCov(X, Y)$

3.5.2 Correlation Coefficient

$$Cor(X, Y) = \rho_{X,Y} = \frac{Cov(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}}$$

- $-1 \leq \rho_{X,Y} \leq 1$
- $\rho_{X,Y}$ = measure of degree of **linear** r/s b/w X and Y
- If X, Y are independent, then $\rho_{X,Y} = 0$.

4 Special Probability Distributions

4.1 Discrete Uniform Distribution

If RV X assumes the values $x_1, x_2, \dots, x_k$ with equal probability, then X has a discrete uniform distribution, and the probability function is $f_X(x) = \frac{1}{k}, x = x_1, x_2, \dots, x_k$, and 0 otherwise.

4.1.1 Mean and Variance of Discrete Uniform Distribution

$$\begin{aligned} \mu &= E(X) = \sum x f_X(x) = \frac{1}{k} \sum_{i=1}^k x_i \\ \sigma^2 &= V(X) = \sum (x - \mu)^2 f_X(x) = \frac{1}{k} \sum_{i=1}^k (x_i - \mu)^2 \\ \sigma^2 &= E(X^2) - \mu^2 = \frac{1}{k} (\sum_{i=1}^k x_i^2) - \mu^2 \end{aligned}$$

4.2 Bernoulli and Binomial Distribution

The collection of all probability distributions for different values of the param is called a **family** of probability distributions.

4.2.1 Bernoulli Distribution

- A random experiment with only 2 possible outcomes.
- RV X has a Bernoulli distribution if the probability function of X is $f_X(s) = p^x(1-p)^{1-x}, x = 0, 1$ where $0 < p < 1$, 0 for other X values. p is the param.
- $\Pr(X = 1) = p$ and $\Pr(X = 0) = 1 - p = q$
- $\mu = E(X) = p, \sigma^2 = V(X) = p(1-p) = pq$

4.2.2 Binomial Distributions

- RV X has a **Binomial** distr with 2 params n and p ($X \sim B(n, p)$), if the prob fn of X is $\Pr(X = x) = f_X(x) = \binom{n}{x} p^x q^{n-x}$ for $x = 0, 1, \dots, n$ where $0 < p < 1$
- X is the no of successes in n independent Bernoulli trials.
- Bernoulli distribution is a special case of Binom distr when $n = 1$
- $\mu = E(X) = np, \sigma^2 = V(X) = npq$
- Conditions: (1) consists of n repeated Bernoulli trials, (2) Only 2 possible outcomes in each trial, (3) $\Pr(\text{success}) = p$ is constant in each trial, (4) trials are **independent**

4.2.3 Negative Binomial Distribution

- Like binom, but trials will be repeated until a **fixed** no of successes occur (prob the k -th success occurs on the x -th trials vs prob x successes in n trials)
- Let X be a RV represents no of trials to produce k successes in a sequence of independent Bernoulli trials, $B \sim NB(k, p)$
- $\Pr(X = x) = f_X(x) = \binom{x-1}{k-1} p^k q^{x-k}$ for $x = k, k+1, k+2, \dots$
- $E(X) = \frac{k}{p}, V(X) = \frac{(1-p)k}{p^2}$
- Special case: No of trials to the first success is **Geometric** distribution ($X \sim NB(1, p) \equiv X \sim \text{Geom}(p)$)

4.3 Poisson Distribution

- RV X , no of successes during a given time interval/in a specified region
- $\Pr(X = x) = f_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, 3, \dots$ where λ = average no of successes occurring in the given time interval/specified region
- $E(X) = V(X) = \lambda$
- Properties: (1) No of successes in one time interval/specified region are **independent** of those in any other disjoint time interval/region of space, (2) The prob of a single success during a short time interval/in a small region is **proportional** to length of time interval/size of region, and does not depend on no of successes outside this time interval/region, (3) The prob of more than one success in such a short time interval/falling in such a small region is **negligible**

4.4 Poisson Approximation to the Binomial Distribution

- Let $X \sim B(n, p)$, suppose that $n \rightarrow \infty$ and $p \rightarrow 0$ such that $\lambda = np$ remains a constant as $n \rightarrow \infty$, then X will have approx a Poisson distr with param np
- $\lim_{p \rightarrow 0, n \rightarrow \infty} \Pr(X = x) = \frac{e^{-np} (np)^x}{x!}$
- If $p \rightarrow 1$, can stil use by swapping success & failure s.t. $p \rightarrow 0$

4.5 Continuous Uniform Distribution (U) or Rectangular Distribution

- RV has **uniform** distr over interval $[a, b]$, $-\infty < a < b < \infty$, denoted by $U(a, b)$ if its pdf is $f_X(x) = \frac{1}{b-a}$ for $a \leq x \leq b$ and 0 otherwise.
- $E(X) = \frac{a+b}{2}, V(X) = \frac{1}{12}(b-a)^2$

4.6 Exponential Distribution

- Continuous RV X assuming all non-negative values has an exponential distr with param $a > 0$ if its pdf is $f_X(x) = ae^{-ax}$ for $x > 0$ and 0 otherwise.
- $E(X) = \frac{1}{a}, V(X) = \frac{1}{a^2}, \int_{-\infty}^{\infty} f(x) dx = 1$
- pdf $f_X(x) = \frac{1}{\mu} e^{-x/\mu}$ for $x > 0$ and 0 otherwise. Then $E(X) = \mu, V(X) = \mu^2$
- A model for the distribution of times between the occurrence of successive events (eg. customers arriving at a facility/calls coming in to a switchboard)

4.6.1 No Memory Property of Exponential Distribution

Suppose $X \sim \text{Exp}(\alpha)$ where $\alpha > 0$, then for any 2 positive numbers s and t , $\Pr(X > s + t \mid X > s) = \Pr(X > t)$

4.7 Normal Distribution (Gaussian Distribution)

RV X assuming all real values has a normal distribution if its pdf is $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp(-\frac{(x-\mu)^2}{2\sigma^2})$ where $-\infty < x < \infty$ and $\sigma > 0$

4.7.1 Properties

- Graph of the distribution is bell-shaped and symmetrical about the vertical line $x = \mu$ ($\Pr(Z \geq z_\alpha) = \Pr(Z \leq -z_\alpha) = \alpha$)
- Max point occurs at $x = \mu$, its value is $\frac{1}{\sqrt{2\pi}\sigma}$
- $E(X) = \mu$, $V(X) = \sigma^2$
- Total area under the curve and above the horizontal axis is 1.
- The normal curve approaches the horizontal axis asymptotically in either direction away from mean. 2 normal curves are identical in shape with same σ^2 , but centered around their means. As σ increases, the curve flattens; as σ decreases, the curve sharpens.
- If $X \sim N(\mu, \sigma^2)$ and $Z = \frac{X-\mu}{\sigma}$, then $Z \sim N(0,1)$

4.8 Normal Approximation to the Binomial Distribution

When $np > 5$ and $nq > 5$ ($n \rightarrow \infty, p \rightarrow \frac{1}{2}$)
 If $X \sim B(np, npq)$, then as $n \rightarrow \infty$, $Z = \frac{\bar{X}-np}{\sqrt{npq}}$ is approx. $\sim N(0,1)$

4.8.1 Continuity Correction

- $\Pr(X = k) \approx \Pr(k - \frac{1}{2} < X < k + \frac{1}{2})$
- $\Pr(a \leq X \leq b) \approx \Pr(a - \frac{1}{2} < X < b + \frac{1}{2})$, $\Pr(a < X \leq b) \approx \Pr(a + \frac{1}{2} < X < b + \frac{1}{2})$
 $\Pr(a \leq X < b) \approx \Pr(a - \frac{1}{2} < X < b - \frac{1}{2})$, $\Pr(a < X < b) \approx \Pr(a + \frac{1}{2} < X < b - \frac{1}{2})$
- $\Pr(X \leq c) = \Pr(0 \leq X \leq c) \approx \Pr(-\frac{1}{2} < X < c + \frac{1}{2})$
- $\Pr(X > c) = \Pr(c < X \leq n) \approx \Pr(c + \frac{1}{2} < X < n + \frac{1}{2})$

5 Sampling and Sampling Distributions

5.1 Population and Sample

- Totality of all possible outcomes is called **popn**
- A **sample** is any subset of a popn
- 2 kinds of popn: (1) **Finite** popn, consisting of a finite number of elements, (2) **Infinite** popn, consisting of an infinitely (countable and uncountable) large number of elements
- A set of n observations from a given popn is a **sample** of size n
- Each observation in popn can be considered as a value of a RV with pdf $f_X(x)$

5.2 Random Sampling

5.2.1 Simple Random Sampling

A SRS of n observations is a sample chosen in such a way that **every subset** of n observations of the popn has the **same probability** of being selected

5.2.2 Sampling from a Finite Population

- Sampling without replacement:** There are $\binom{N}{n}$ samples of size n to be drawn from popn of size N without replacement. Each sample has the same probability of $1/(\binom{N}{n})$ of being selected.
- Sampling with replacement:** There are N^n samples of size n drawn from popn of size N . Each sample has the same probability $\frac{1}{N^n}$ of being selected.

5.2.3 Sampling from an Infinite Population

Random if (1) In each draw all elements of the popn have the **same probability of being selected**, (2) Successive draws are **independent**

5.2.4 Theorem

Let X be an RV with pdf $f_X(x)$, $X_1, X_2, ..., X_n$ be n independent RV each having the same distr as X . Then $(X_1, X_2, ..., X_n)$ is called a **random sample** of size N from a popn with distribution $f_X(x)$. The joint pdf is $f_{X_1, X_2, ..., X_n}(x_1, x_2, ..., x_n) = f_{X_1}(x_1) f_{X_2}(x_2) ... f_{X_n}(x_n)$

5.3 Sampling distribution of sample mean ($\bar{X}$)

5.3.1 Statistic and Sampling Distribution

Sampling distribution = probability distribution of a statistic

5.3.2 Sample Mean

$X_1, X_2, ..., X_n$ is a random sample of size $n \Rightarrow$ sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

5.3.3 Theorem

For random samples of size n taken from **infinite** popn or from **finite popn with replacement** having popn mean μ and popn std dev σ , $\bar{X}$ has its mean and std dev: $\mu_{\bar{X}} = \mu_X$ and $\mu_{\bar{X}^2} = \frac{\sigma_X^2}{n}$, i.e. $E(\bar{X}) = E(X)$ and $V(\bar{X}) = \frac{V(X)}{n}$

5.3.4 Law of Large Number (LLN)

Let $X_1, X_2, ..., X_n$ be a random sample of size n from a popn having any distribution with mean μ and **finite** popn variance σ^2 . Then for any $\epsilon \in \mathbb{R}$, $\Pr(|\bar{X} - \mu| > \epsilon) \rightarrow 0$ as $n \rightarrow \infty$ (as sampe size increases, probability that sample and popn mean differs goes to 0)

5.4 Central Limit Theorem

Let $X_1, X_2, ..., X_n$ be a random sample of size n , popn mean μ , finite popn var σ^2 . $\bar{X}$ is **approx. normal** with mean μ and variance $\frac{\sigma^2}{n}$ if n is sufficiently large. Hence, $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ follows approx. $N(0,1)$

5.4.1 Theorem

If $X_i, i = 1, 2, ..., n$ are $N(\mu, \sigma^2)$, then $\bar{X}$ is $n(\mu, \frac{\sigma^2}{n})$ regardless of the sample size n . (Same thing if approximately follow)

5.5 Sampling Distribution of the Difference of 2 Sample Means

5.5.1 Theorem

If independent samples of sizes n_1 and n_2 (each ≥ 30) from 2 large/infinite popns, disc or cont, with means μ_1, μ_2 , var σ_1^2, σ_2^2 , then the sampling distr of the differences of means $\bar{X}_1$ and $\bar{X}_2$ is approx. normally distr with $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$ and $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ is $Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$ approx. $\sim N(0,1)$

5.6 Chi-square distribution

If Y is RV with pdf $f_Y(y) = \frac{1}{2^{n/2}\Gamma(n/2)} y^{n/2-1} e^{-y/2}$ for $y > 0$ and 0 otherwise, then Y has a **chi-square distribution with n degrees of freedom**, denoted $\chi^2(n)$, where $n \in \mathbb{Z}^+$ and $\Gamma(\cdot)$ is the gamma function

5.6.1 Gamma function

$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx = (n-1)!$ for $n \in \mathbb{Z}^+$

5.6.2 Properties

- If $Y \sim \chi^2(n)$, then $E(Y) = n$ and $V(Y) = 2n$
- For large n , $\chi^2(n)$ approx. $\sim N(n, 2n)$
- If $Y_1, Y_2, ..., Y_k$ are **independent** chi-square RV with $n_1, n_2, ..., n_k$ degrees of freedom, then $Y_1 + Y_2 + ... + Y_k$ has a chi-square distr with $n_1 + n_2 + ... + n_k$ degrees of freedom: $\sum_{i=1}^k Y_i \sim \chi^2(\sum_{i=1}^k n_i)$
- $\Pr(Y \geq \chi^2(n; \alpha)) = \alpha$ where $Y \sim \chi^2(n)$. $\Pr(Y \leq \chi^2(n; 1 - \alpha)) = \alpha$

5.6.3 Theorem

- $X \sim N(0,1) \Rightarrow X^2 \sim \chi^2(1)$. $X \sim N(\mu, \sigma^2) \Rightarrow (\frac{X-\mu}{\sigma})^2 \sim \chi^2(1)$
- Let $X_1, X_2, ..., X_n$ be a random sample from a normal popn with mean μ and var σ^2 . Define $Y = \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2}$, then $Y \sim \chi^2(n)$

5.6.4 The sampling distribution of $\frac{(n-1)S^2}{\sigma^2}$

Let $X_1, X_2, ..., X_n$ be a random sample from a popn, then $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the **sample variance**

Theorem: If S^2 is the sample variance, then $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$

5.7 The t-distribution

Let Z be standard normal variable and U a chi-square RV with n degs of freedom. If Z and U are independent, and let $T = \frac{Z}{\sqrt{U/n}}$, then the RV T follows the t-distribution with n degs of freedom

5.7.1 The pdf of a t-distribution

$f_T(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{n\pi}\Gamma(\frac{n}{2})} (1 + t^2 \frac{n+1}{n})^{-\frac{n+1}{2}}, -\infty < t < \infty$

5.7.2 Properties

- Graph of t-distribution is symmetric about the vertical axis, resembles standard normal distr
- pdf of t-distr $\rightarrow$ pdf of std normal distr when $n \rightarrow \infty$
- $E(T) = 0$ and $V(T) = \frac{n}{n-2}$ for $n > 2$
- Remark: if the random sample was selected from a normal popn, then $Z \sim N(0,1)$ and $U \sim \chi^2(n-1)$, then $T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$

5.8 The F-distribution

Let $U \sim \chi^2(n_1)$ and $V \sim \chi^2(n_2)$, then $F = \frac{U/n_1}{V/n_2}$ is called **F-distribution** with (n_1, n_2) degs of freedom.

5.8.1 pdf

$f_F(x) = \frac{n_1^{n_1/2} n_2^{n_2/2} \Gamma(\frac{n_1+n_2}{2}) x^{n_1/2-1}}{\Gamma(\frac{n_1}{2}) \Gamma(\frac{n_2}{2}) (n_1 x + n_2)^{(n_1+n_2)/2}}$ for $x > 0$ and 0 otherwise.

5.8.2 Theorem

$F \sim F(n, m) \rightarrow \frac{1}{F} \sim F(m, n)$ and $F(n_1, n_2; 1 - \alpha) = \frac{1}{F(n_2, n_1; \alpha)}$

6 Estimation based on Normal Distribution

Some characteristics of elements in a popn can be represented by an RV X with pdf $f_X(x; \theta)$ where the form is assumed known, values of random sample can be observed except unknown param θ

6.1 Point Estimation of Mean and Variance

- Point estimator** = stat $\hat{\theta} = \hat{\Theta}(X_1, X_2, ..., X_n)$ to estimate unknown param θ
- A **statistic** is a function of the random sample with does not depend on any unknown param. (e.g. sum/max of observations)
- An **estimator** is the statistic used to obtain a point estimate. ($\bar{X}$ is an estimator of μ , the value of $\bar{X}$, $\bar{x}$ is an estimate of μ)

6.1.1 Unbiased Estimator

- A statistic $\hat{\Theta}$ is an **unbiased estimator** of the param θ if $E(\hat{\Theta}) = \theta$
- $\bar{X}$ is an unbiased estimator of μ
- $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is unbiased estimator of σ^2 , i.e. $E(S^2) = \sigma^2$

6.2 Interval Estimation

- Form: $\hat{\theta}_L < \theta < \hat{\theta}_U$ where $\hat{\theta}_L$ and $\hat{\theta}_U$ depend on (1) value of the stat $\hat{\Theta}$ for a particular sample, (2) the sampling distribution of $\hat{\Theta}$
- $\hat{\theta}_L$ and $\hat{\theta}_U$ = lower and upper confidence limit, $\hat{\theta}$ = point estimate
- Seek a random interval s.t. $\Pr(\hat{\Theta}_L < \theta < \hat{\Theta}_U) = 1 - \alpha$
- The interval computed from the selected sample is $(1 - \alpha)100\%$ confidence interval for α
- $(1 - \alpha)$ is **confidence coefficient** or **degree of confidence**

6.3 Confidence Intervals for the Mean

6.3.1 Known Variance Case

- With (i) known variance, (ii) the population is normal, or $n \geq 30$
- $\Pr(-z_{\alpha/2} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}) = \Pr(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$
- If $\bar{X}$ is sample mean from a popn with variance σ^2 , a $(1 - \alpha)100\%$ confidence interval for μ is as expression inside Pr above (middle)

Sample size for Estimating μ
 For a given margin of error e , **sample size** is $n \geq (z_{\alpha/2} \frac{\sigma}{e})^2$

6.3.2 Unknown Variance Case

- With (i) unknown popn variance, (ii) the popn is normal or very close to, (iii) sample size is small ($n < 30$)
- Let $T = \frac{(\bar{X}-\mu)}{S/\sqrt{n}}$ where S^2 is sample variance, then $T \sim t_{n-1}$.
- $\Pr(-t_{n-1; \alpha/2} < T < t_{n-1; \alpha/2}) = \Pr(\bar{X} - t_{n-1; \alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{n-1; \alpha/2} \frac{S}{\sqrt{n}}) = 1 - \alpha$
- $\bar{X}$ and S are sample mean and std dev, a $(1 - \alpha)100\%$ confidence interval for μ is as expr inside Pr above (middle)
- For large $n > 30$, the t-distr approx. $N(0,1)$. Hence the conf interval is replace $t_{n-1; \alpha/2}$ with $z_{\alpha/2}$

6.4 Confidence Intervals for the Difference between 2 Means

2 popns with means μ_1, μ_2 , then $\bar{X}_1 - \bar{X}_2$ is the point estimator of $\mu_1 - \mu_2$

6.4.1 Known variance

- When $\sigma_1^2 \neq \sigma_2^2$ and (2 popns are normal or n_1, n_2 both ≥ 30)

$$(\bar{X}_1 - \bar{X}_2) \sim N(\mu_1 - \mu_2, \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2})$$
$$\Pr\left(-z_{\alpha/2} < \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}} < z_{\alpha/2}\right) = 1 - \alpha$$

- $(1 - \alpha)$ 100% confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{X}_1 - \bar{X}_2) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{X}_1 - \bar{X}_2) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

6.4.2 Large sample Confidence Interval for Unknown Variances

σ_1^2, σ_2^2 are unknown, n_1, n_2 both ≥ 30 , replace σ_1^2, σ_2^2 by their estimates S_1^2, S_2^2

6.4.3 Unknown but Equal Variances

- $\sigma_1^2 = \sigma_2^2$, 2 popns are normal, n_1, n_2 both ≤ 30

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \sim \chi^2_{n_1 + n_2 - 2}$$
$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1 + n_2 - 2}$$

- $(1 - \alpha)$ 100% confidence interval for $\mu_1 - \mu_2$ is

$$(\bar{X}_1 - \bar{X}_2) - t_{n_1 + n_2 - 2; \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{X}_1 - \bar{X}_2) + t_{n_1 + n_2 - 2; \alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

6.4.4 Unknown but Equal Variances for Large Samples

For n_1, n_2 both ≥ 30 , replace $t_{n_1 + n_2 - 2; \alpha/2}$ by $z_{\alpha/2}$

6.4.5 C.I. for the Difference between 2 Means for Paired (Dependent) Data

- E.g. same individual before and after (related observations)

$$\text{Point estimate of } \mu_D = \mu_1 - \mu_2 \text{ is given by } \bar{d} = \frac{1}{n} \sum_{i=1}^n d_i = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)$$
$$\text{Pt est of } \sigma_D^2 \text{ is given by } s_D^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2 = \frac{1}{n-1} (\sum_{i=1}^n d_i^2 - n\bar{d}^2)$$

- Small sample, approximate normal popn:**

$$T = \frac{\bar{d} - \mu_D}{s_d / \sqrt{n}} \sim t_{n-1}$$
$$(1 - \alpha)100\% \text{ CI for } \mu_D = \bar{d} - t_{n-1; \alpha/2} \left(\frac{S_D}{\sqrt{n}}\right) < \mu_D < \bar{d} + t_{n-1; \alpha/2} \left(\frac{S_D}{\sqrt{n}}\right)$$

- For large sample ($n > 30$)**, replace $t_{n-1; \alpha/2}$ by $z_{\alpha/2}$

6.5 C.I. for Variances and Ratio of Variances

C.I. for σ or σ_1/σ_2 is obtained by taking square root of each end point

6.5.1 C.I. for a Variance of a Normal Popn

Sample var $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} (\sum_{i=1}^n X_i^2 - n\bar{X}^2)$ is pt est of σ^2

$$\mu \text{ is known: } \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{n; \alpha/2}} < \sigma^2 < \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi^2_{n; 1 - \alpha/2}}$$
$$\mu \text{ is unknown: } \frac{(n-1)S^2}{\chi^2_{n-1; \alpha/2}} < \sigma^2 < \frac{(n-1)S^2}{\chi^2_{n-1; 1 - \alpha/2}}$$

6.5.2 C.I. for the Ratio of 2 Variances of Normal Popn with Unknown Means

$$\frac{S_1^2}{S_2^2} \frac{1}{F_{n_1-1, n_2-1; \alpha/2}} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{S_1^2}{S_2^2} F_{n_2-1, n_1-1; \alpha/2}$$

7 Hypotheses Testing based on Normal Distribution

7.1 Null and Alternative Hypotheses

H_0 : formulate with the hope of rejecting, which leads to acceptance of H_1

7.1.1 Types of errors

- Type I** (serious): $\Pr(\text{Reject } H_0 \mid H_0 \text{ is true}) = \alpha = \text{level of significance}$
- Type II**: $\Pr(\text{Do not reject } H_0 \mid H_0 \text{ is false}) = \beta$, Power of a test = $1 - \beta$

7.1.2 Acceptance and Rejection Regions

Rejection (critical) region and acceptance region are separated by **critical value**

7.2 Hypotheses Testing Concerning Mean

7.2.1 Hypo Testing on Mean with Known Variance

Variance σ^2 is known and underlying distr is normal or $n > 30$

Two-sided test:

- Test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$. Under H_0 , we have $\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$
- $\bar{x}_1 < \bar{X} < \bar{x}_2$ or $-z_{\alpha/2} < Z < z_{\alpha/2}$ defines acceptance region.
- $\bar{x}_1, \bar{x}_2 = \mu_0 \mp z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ or $z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
- If $\bar{X}$ falls in acceptance region, conclude $\mu = \mu_0$. Else reject H_0 & accept H_1
- Equivalent to finding $(1 - \alpha)$ 100% C.I. for μ : accept H_0 if C.I. covers μ_0

One-sided test: Test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$ or $H_1 : \mu < \mu_0$. The rest is the same.

7.2.2 p-value Approach to Testing (observed level of significance)

- p-value:** Prob of obtaining a test stat more extreme ($\leq$ or $\geq$) than the observed sample given H_0 is true. **Compare sample mean $\bar{X}$ to Z**
- $p\text{-value} < \alpha$, reject H_0 ; $p\text{-value} \geq \alpha$, do not reject H_0

7.2.3 Hypo Testing on Mean with Unknown Variance

Variance unknown and underlying distr is normal

Let $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ where S^2 is sample variance. Reject H_0 if in rejection region

7.3 Hypo Testing Concerning Difference Between 2 Means

7.3.1 Known Variances

Known variances, norm distr, or n_1, n_2 both ≥ 30 , use section 6.4.1

7.3.2 Large Sample Testing with Unknown Variances

Unknown variances, both n_1, n_2 both ≥ 30 , use section 6.4.2

7.3.3 Unknown but Equal Variances

$\sigma_1^2 = \sigma_2^2$, norm distr, n_1, n_2 both ≤ 30 , use section 6.4.3

7.3.4 Paired Data

Use section 6.4.5

7.4 Hypo Testing Concerning Variance

7.4.1 One Variance Case

- Assume normal distribution where σ^2 is unknown.
- $H_0 : \sigma^2 = \sigma_0^2$, use test stat: $\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$
- Reject H_0 if within critical region:

H_1	Critical Region
$\sigma^2 > \sigma_0^2$	$\chi^2 > \chi^2_{n-1; \alpha}$
$\sigma^2 < \sigma_0^2$	$\chi^2 < \chi^2_{n-1; 1-\alpha}$
$\sigma^2 \neq \sigma_0^2$	$\chi^2 < \chi^2_{n-1; 1-\alpha/2}$ or $\chi^2 > \chi^2_{n-1; \alpha/2}$

7.4.2 Hypo Testing Concerning Ratio of Variances

- Assume normal distr, unknown mean.
- $H_0 : \sigma_1^2 = \sigma_2^2$, use test stat: $F = \frac{S_1^2}{S_2^2}$
- Reject H_0 if within critical region:

H_1	Critical Region
$\sigma_1^2 > \sigma_2^2$	$F > F_{n_1-1, n_2-1; \alpha}$
$\sigma_1^2 < \sigma_2^2$	$F < F_{n_1-1, n_2-1; \alpha}$
$\sigma_1^2 \neq \sigma_2^2$	$F > F_{n_1-1, n_2-1; 1-\alpha/2}$ or $F > F_{n_1-1, n_2-1; \alpha/2}$

8 Extra Notes

Small letter distribution are inverse